SatQuery AI

Website User Guide — Earth View & Analysis Workspace

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This guide walks through the SatQuery AI web application in the order you would actually use it: launch, read the screen, navigate the globe, explore imagery layers, open mission sites, analyse a region in the workspace, ask the AI questions, play time lapses, and export session results. Follow it top to bottom for a first tour, or jump to any numbered section.

GETTING STARTED

Launching the app

Option	Steps
One-click (recommended)	Double-click `START_DEMO.bat` in the project root. It starts the AI backend on port 7860 and opens the app in your default browser.
Manual	Start the backend: `python -m uvicorn satquery.api:app --port 7860`, then open `docs/index.html` through a local server or file URL.
View-only	Just open `docs/index.html` in any browser. The globe, imagery and dossiers work without the backend; only live AI answers need it.

On load you see a brief splash, then the live Earth globe. The header line shows which satellite feed is active and the date it was captured, e.g. VIIRS SNPP, captured 2026-09-08 (latest available). That suffix means imagery over some regions stepped back a day because NASA had not finished ingesting today's passes there — this is intentional and guarantees you never see blacked-out hemispheres. The globe needs an internet connection (NASA GIBS + Esri imagery are streamed live).

READING THE SCREEN

Region	What it is
Header (top-left)	SatQuery AI title, the active satellite feed + captured date, a GIBS LIVE badge, and three buttons: Insights, Docs & API, and Keyboard shortcuts.
Search pill (top-centre)	One box that finds places, mission sites, or raw coordinates like `26.14, 91.73`. Press Enter to fly there; arrow keys pick a suggestion.
Missions panel (left)	Category chips (All, Flood, Urban, Coastal, Agriculture, Forest, Wildfire, Glacier) and the nine mission-site cards. Click a card to preview; it opens the site dossier popup.
Imagery panel (right)	The eight base imagery layers (keys 1–8), the analytic overlay stack with an "+ Add overlay" menu, and context toggles (borders, place labels, graticule, live activity).

Region	What it is
Status bar (bottom)	Cursor lat/lon, current view, session counters (questions asked, sites analysed, 5-capability coverage) and frame rate.

Table 1 — The five permanent screen regions.

NAVIGATING THE GLOBE

Navigation follows three levels of detail, each with its own imagery source:

Level	You see	How to get there
1 — Globe	Whole-Earth view, NASA daily satellite mosaics (VIIRS / MODIS, ~250–375 m per pixel).	Drag to orbit. Scroll to zoom toward the cursor.
2 — Pass view	A flat high-resolution regional pass of wherever you zoomed.	Keep zooming in — the viewer hands off automatically around a few hundred km. `Shift + click` inside it for even higher resolution.
3 — Aerial	Sub-metre aerial imagery (Esri) at street / building scale.	Zoom under roughly 60 km width; the handoff is seamless and automatic.

Table 2 — The three zoom levels.

Hold `Shift` while scrolling for a strong 3× zoom. Press `Esc` anywhere to back out one level (aerial → pass → globe → close dialogs). `Shift + click` or right-click any point on Earth to jump straight to the analysis workspace at that location.

FINDING A PLACE

Click the search pill and start typing. It matches three things: place names, the nine mission sites, and raw coordinates. For example typing `assam` or `26.14, 91.73` both take you to the Assam flood plain. Use the up/down arrows to walk the suggestion list and Enter to fly; Esc dismisses it. The globe animates to the selected location.

CHOOSING IMAGERY

Base layers (keys 1–8)

Key	Layer	Best for
1	VIIRS SNPP true color	Default. Daily global natural-color imagery; storms, smoke, snow, sea ice.
2	VIIRS NOAA-20 true color	Second daily pass when SNPP is cloudy or late.

Key	Layer	Best for
3	MODIS Terra true color	Long-archive morning pass; same-day VIIRS fills equatorial gaps.
4	MODIS Aqua true color	Afternoon pass; compare cloud cover against Terra.
5	VIIRS SWIR false color	Burn scars, floods, snow/ice vs cloud separation.
6	VIIRS cloud/snow false color	Separating snow and ice from cloud.
7	Black Marble night lights	City footprints and infrastructure at night (2016 composite).
8	VIIRS N20 SWIR false color	Fire and flood contrast on the alternate VIIRS satellite.

Table 3 — Base imagery layers, switchable with number keys 1–8 or the Imagery panel.

Overlays and context

Click "+ Add overlay" in the Imagery panel to stack translucent analytic layers on top of the base: aerosol optical depth (smoke/dust plumes), cloud-top temperature, precipitable water, sea-surface temperature, chlorophyll-a, snow cover, sea ice, and active-fire detections from VIIRS and MODIS. Overlays are meant to be read against a true-color base — e.g. fire dots over smoke. The Context toggles add reference geography: country borders (`B`), place labels for 76,000+ cities, towns and villages (`L`), the lat/lon graticule (`G`), and live event markers from USGS earthquakes and NASA EONET (`A`).

MISSION SITES AND DOSSIERS

The nine mission sites are curated analysis locations: Assam Flood Plain, Delhi Urban Sprawl, Sundarbans Delta, Punjab Agri Belt, Gangotri Glacier, Rondônia Deforestation, Sierra Nevada Fire Scar, Nile Delta, and Dubai Coast. Filter them with the category chips. Clicking a site card flies the globe there and opens a popup with its story; from the popup, **Open workspace** enters the full analysis workspace and **Refocus** just re-centres the view. The site markers on the globe itself are also clickable.

THE ANALYSIS WORKSPACE

Open the workspace in any of three ways: click a mission site and choose **Open workspace**; `Shift + click` anywhere on Earth; or right-click a point. It is a large window with the imagery viewer on the left and three tabs on the right.

Control	What it does
Date picker	Choose which day's imagery to display (steps back automatically when the newest day has no coverage there).
Layer selector	Swap the imagery layer for this window independently of the globe.
Time lapse	Opens the playback modal (section 9).
Info	Shows imagery metadata: source, resolution, date, coverage.
Download frame	Saves the current imagery frame as an image file.

Control	What it does
Google Maps	Opens the same coordinates in Google Maps for cross-reference.
Viewer canvas	Drag to pan, scroll to zoom, shift-click to recenter the window on a point.

Table 4 — Workspace viewer controls.

The footer under the viewer shows coordinates, window span, resolution and imagery date; a scale bar sits at the right. Answers from the AI draw gold dashed boxes directly on this canvas.

The three tabs

Tab	Contents
Dossier	The site's analysis brief: the problem, approach, KPIs, an event timeline, the sensors used, findings with confidence values, and recommended actions.
Ask AI	The question box and the agent pipeline (section 8).
Scenes	The raw Sentinel-2 / Sentinel-1 scene metadata behind the analysis pair.

Table 5 — Workspace side tabs.

ASKING THE AI

In the workspace, open the **Ask AI** tab. Type a question about the current window and press Ask (or Enter). Five preset buttons Q1–Q5 cover the canonical query types. As the query runs you watch the six pipeline stages animate — register, fetch, featurise, reason, ground, report — then the answer appears with a confidence value, and any region it refers to is boxed on the imagery. A **trace JSON** button exposes the full audit record, and the model registry shows which head went live for your query.

Ask about…	Example phrasing	Head that answers
What is here	Describe the land cover and major objects visible in this image.	Caption
Where something is	Highlight the water body referred to in the query.	Grounding
What changed	What changed between these two dates, and where?	Change
Change Q&A;	Has the built-up area increased, decreased, or stayed the same?	Change-qa
Sensor fusion	Use the optical and SAR images together to identify built-up and water regions.	Fusion
Anything else	Free-form questions about the window.	VQA

Table 6 — Query types and the head each one activates.

The app runs in two modes. With the backend running it answers from the real agent pipeline (the first query after a cold start is slow while the model loads; later ones are under a second). Without the backend it answers in demonstration mode with the same interface, so the tour never breaks — the header badge and answer text tell you which mode you are in.

TIME LAPSE

From any workspace, click **Time lapse**. Pick 7 or 30 days, then use Play / Prev / Next or the slider to step through daily frames of the same window. The date of the current frame is overlaid on the imagery. Use it to watch floods rise, fires spread, snow retreat or a city expand.

INSIGHTS AND EXPORTS

The Insights button in the header (or `I`) opens session telemetry: every question you asked, which sites and capabilities were covered out of the five prescribed ones, and timing. From here you can export the whole session as JSON or CSV, or reset it. The Docs & API button (`D`) documents the agent pipeline, the model registry and the data sources behind the app.

KEYBOARD SHORTCUTS

Keys	Action
`1 – 8`	Switch base imagery layer
`B`	Toggle country borders
`L`	Toggle place labels (76,000+ places)
`G`	Toggle lat/lon graticule
`A`	Toggle live activity markers (USGS / EONET)
`Shift + scroll`	Strong zoom
`Shift + click`	Open workspace at that point
`right-click`	Open workspace at that point
`drag`	Orbit the globe / pan imagery
`scroll`	Cursor-anchored zoom
`Esc`	Close dialogs / back out one zoom level
`I`	Insights
`D`	Docs & API
`?`	Keyboard shortcuts panel

Table 7 — Complete shortcut list (also available in-app via the ? panel).

TIPS AND TROUBLESHOOTING

Symptom / need	What to do
Header shows an older date with the suffix (latest available)	Expected. NASA ingests daily passes region-by-region over ~20 hours; the app automatically steps back to the newest date with full coverage rather than showing black gaps.
Black or missing imagery in a region	Try another base layer (key 2 or 3) — different satellites ingest at different times. Also check your internet connection; all imagery is streamed live.
First AI answer is slow	The backend lazy-loads its model on the first query (~340 MB). Subsequent queries answer in under a second.
AI answers look canned	The backend is not running. Start it with START_DEMO.bat and reload the page; the app auto-detects it within a couple of seconds.
Water questions abstain on some sites	By design. If there is no water in the window (e.g. the dry Assam upland patch), the system abstains rather than guessing. Try Kaziranga or the Sundarbans for water queries.
A dialog will not close	Press `Esc`. It closes the topmost dialog, or backs out of the pass view.

Table 8 — Quick troubleshooting.

Suggested five-minute tour: open the app → press 5 for SWIR → click the Assam Flood Plain card → Open workspace → press Q3 (change question) → watch the boxes land on the imagery → click Time lapse, 30 days, Play → close with Esc → press I to see your session recorded.